arcsinlem x

$$xff$$

Lemniscate
focal distance: c
half-width: a
relation: $a = c\sqrt{2}$
Cartesian

$$(x^2 + y^2)^2 = a^2(x^2 - y^2)$$

Polar

$$r^2 = a^2 \cos 2\theta$$

Lemniscatic Arcsine

$$\text{arcsl}x = \int_0^x \frac{dt}{\sqrt{1 - t^4}}$$

Angle Addition Formulas

$$\int_a^b x^2 \, dx$$

Circle

Cartesian

$$x^2 + y^2 = a^2$$

Circular Arcsine

$$\arcsin x = \int_0^x \frac{dt}{\sqrt{1 - t^2}}$$